Renewable and Nonrenewable Resources

Chapter 1 Unit 2 Guided Notes · Grade 4 Using Natural Resources for Energy

Name: _____ **Date:** _____

BIG QUESTION **What is the difference between renewable and nonrenewable resources?**

KEY WORDS

renewable resource — a resource that can be restored _____ than it is used up.

nonrenewable resource — a resource that cannot be restored _____ than it is used up.

fossil fuel — a fuel formed from the remains of _____ that lived long ago.

KEY IDEA

KEY IDEA **Abundant does NOT mean renewable.**

Renewable means it comes back as quickly as it is used. Nonrenewable means it does not.

SORT IT

Write each resource in the correct column.

RENEWABLE	NONRENEWABLE
	_____
	_____
	_____

Bank: coal · wood · wind · petroleum · flowing water · natural gas · sunlight · Earth's heat

Availability and Location

Chapter 1 · Page 2

Name: _____ **Date:** _____

RENEWABLE IS NOT ALWAYS AVAILABLE EVERYWHERE

A renewable resource still depends on location and conditions.

Example: In a sunny place, sunlight is renewable and _____ .

Example: In Antarctica in winter, the sun is too _____ to be a dependable energy source.

BENEFITS AND LIMITS

Renewable resources still have costs. Fill in one cost for each.

RENEWABLE RESOURCE	ONE COST OR LIMIT
wind turbines	_____
solar panels	_____
dams for water power	_____

MODEL IT

A resource is renewable if it can be restored as quickly as it is used.

USED SLOWLY → **RESTORED QUICKLY** → **RENEWABLE**

CHECK YOUR THINKING

Coal is plentiful right now, but it forms over millions of years. Is coal renewable or nonrenewable? Explain.

_____

Draw / Label

Chapter 1 · Page 3

Name: _____ **Date:** _____

DRAW / LABEL

Draw one renewable and one nonrenewable resource you use or see.

RENEWABLE	NONRENEWABLE

CHECK YOUR THINKING

Why does it matter how quickly a resource is restored?

CONNECT *Chapter 1 connection: renewable or nonrenewable depends on time — how fast we use it vs. how fast it comes back.*